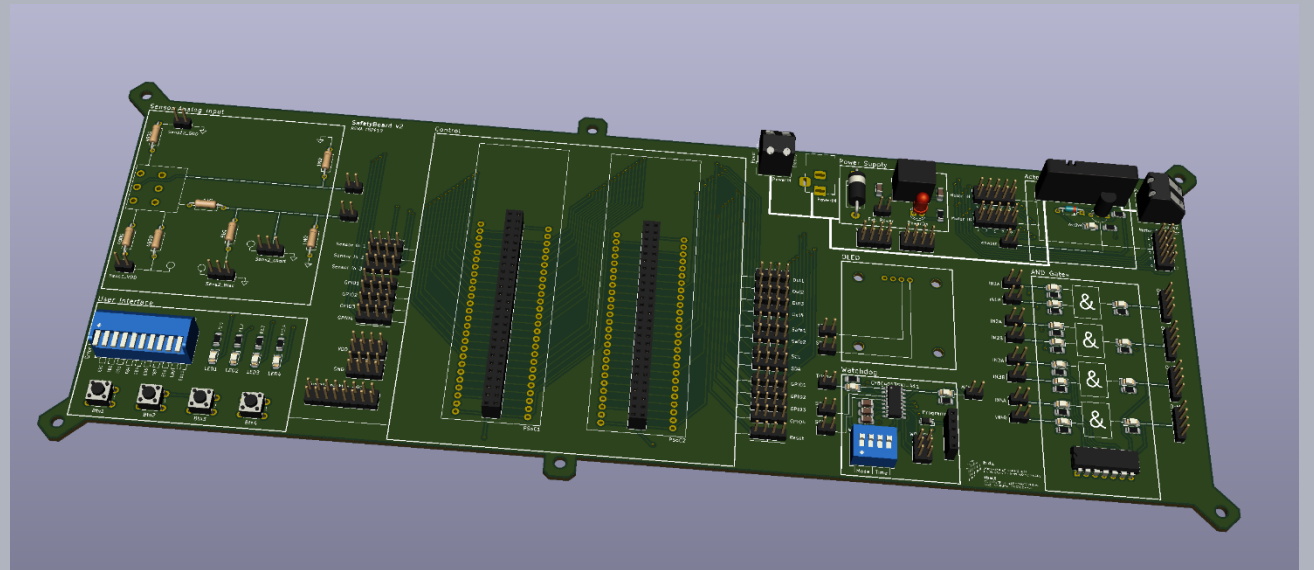


Safety in Embedded Control Systems

Hazard and Risks Analysis

Prof. Dr.-Ing. Peter Fromm



Content

- Item Description
- Hazard and Risk Analysis
- SIL required calculation
- Safety Goals and Safety Requirements

Last Lecture: Risk Mitigation!

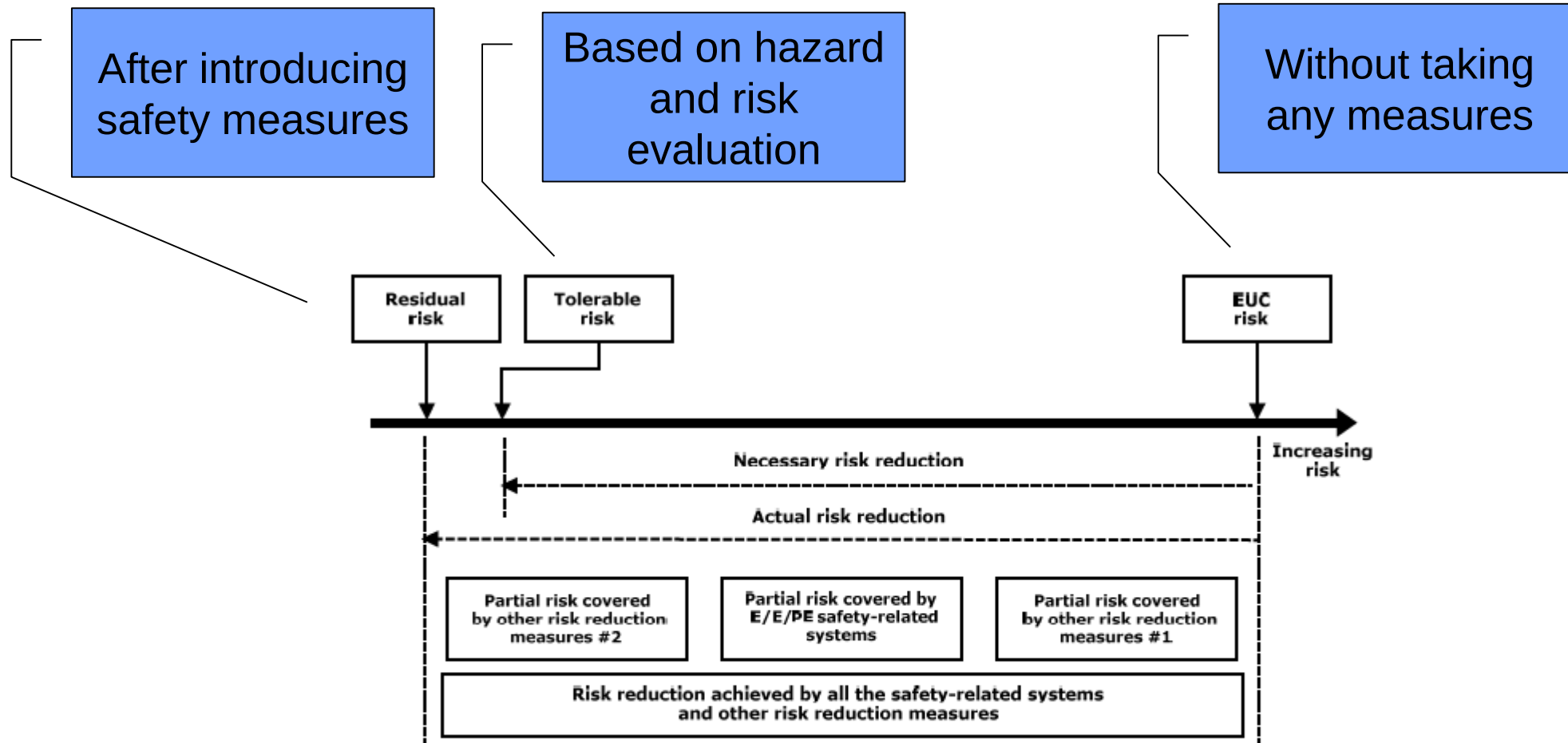
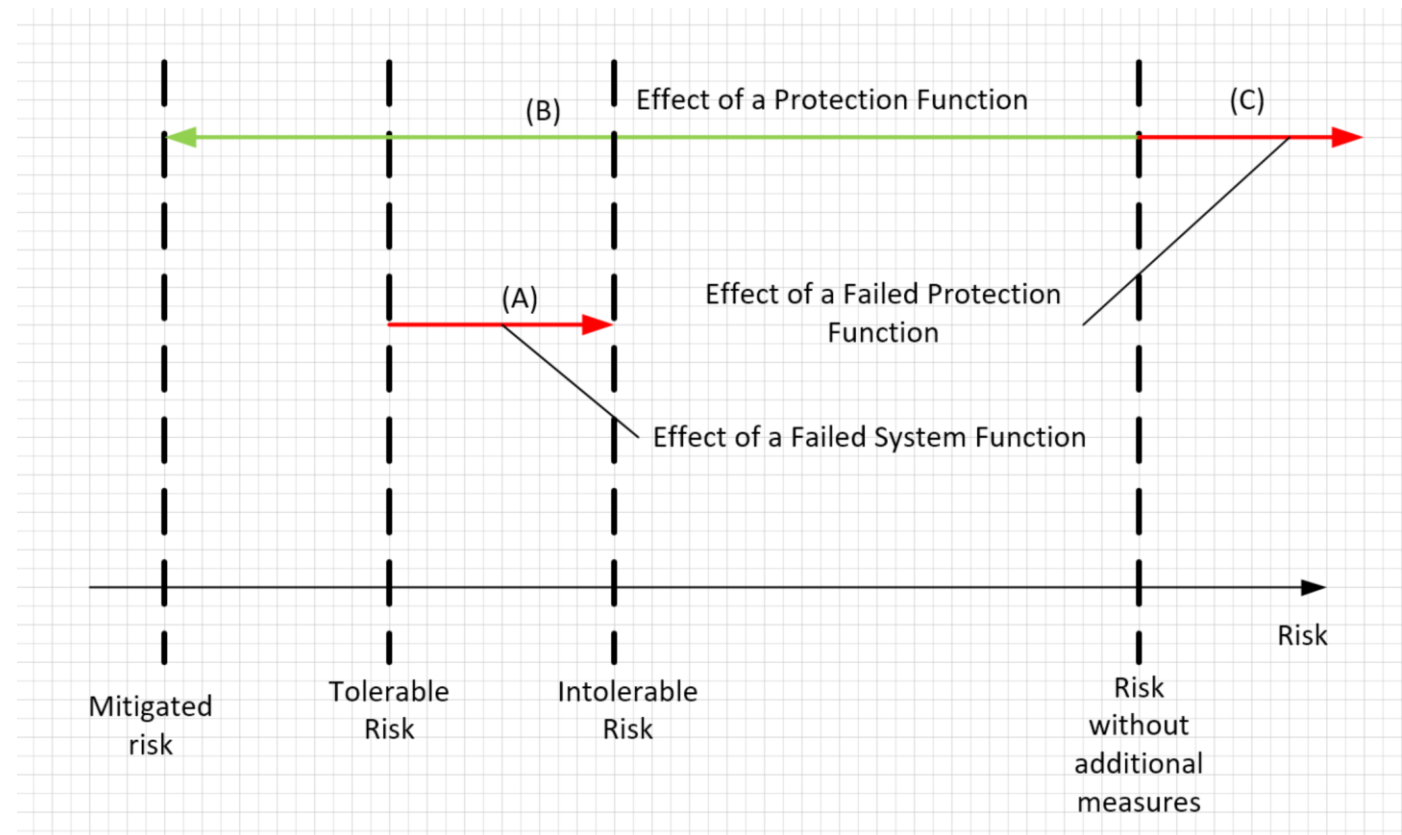


Figure A.1 – Risk reduction – general concepts (low demand mode of operation)

A slightly extended functional view...

- Additional protection functions (e.g. emergency off by a light barrier sensor in front of a dangerous machine) reduce the risk of the overall system
- Safe system functions are required to operate the system in a safe way (e.g. MMI setting the machine speed)
- Both have to be implemented in a safe way in order to avoid harm!



Last Lecture: SIL calculation (61508)

Problem: Very often, a system failure rate is hard to calculate!

E.6 Risk graph example

An example of a risk graph implementation is shown in Figure E.2. Use of the risk parameters C , F , P and W is explained in Table E.1. Each one of these outputs is mapped onto one of the scales (a, b, c, d, e, f, g and h) on these scales (a, b, c, d, e, f, g and h) is an indication of the necessary risk reduction that has to be met by the safety-related system.

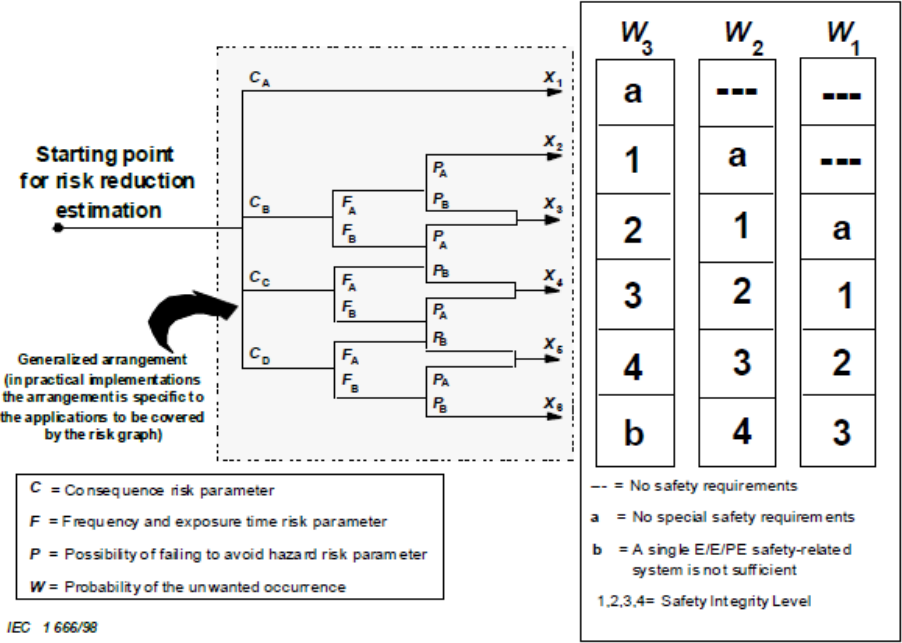


Table 2 – Safety integrity levels – target failure measures for a safety function operating in low demand mode of operation

Safety integrity level (SIL)	Average probability of a dangerous failure on demand of the safety function (PFD_{avg})
4	$\geq 10^{-5}$ to $< 10^{-4}$
3	$\geq 10^{-4}$ to $< 10^{-3}$
2	$\geq 10^{-3}$ to $< 10^{-2}$
1	$\geq 10^{-2}$ to $< 10^{-1}$

e.g. Protection Function

Table 3 – Safety integrity levels – target failure measures for a safety function operating in high demand mode of operation or continuous mode of operation

Safety integrity level (SIL)	Average frequency of a dangerous failure of the safety function [h^{-1}] (PFH)
4	$\geq 10^{-9}$ to $< 10^{-8}$
3	$\geq 10^{-8}$ to $< 10^{-7}$
2	$\geq 10^{-7}$ to $< 10^{-6}$
1	$\geq 10^{-6}$ to $< 10^{-5}$

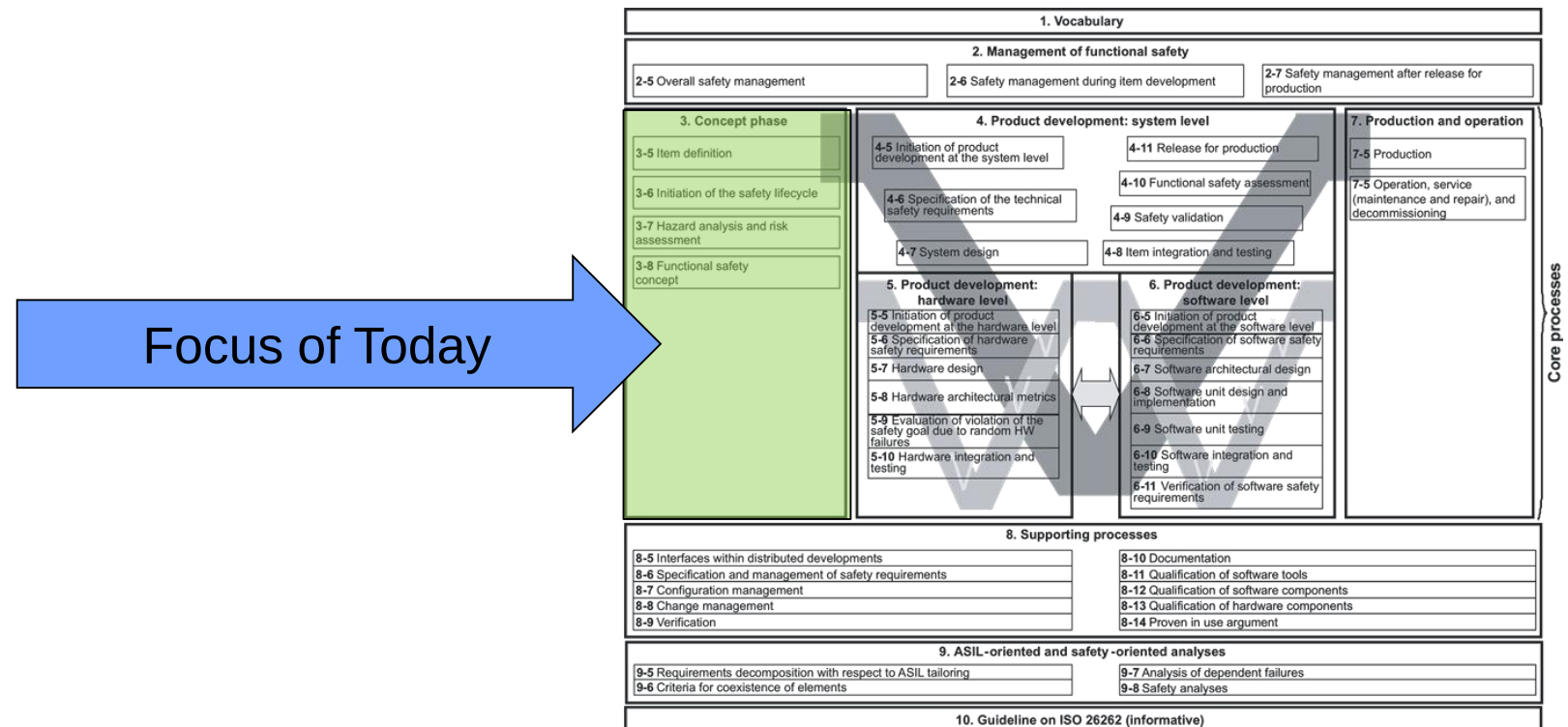
e.g. Safe System Function

criticality

reliability

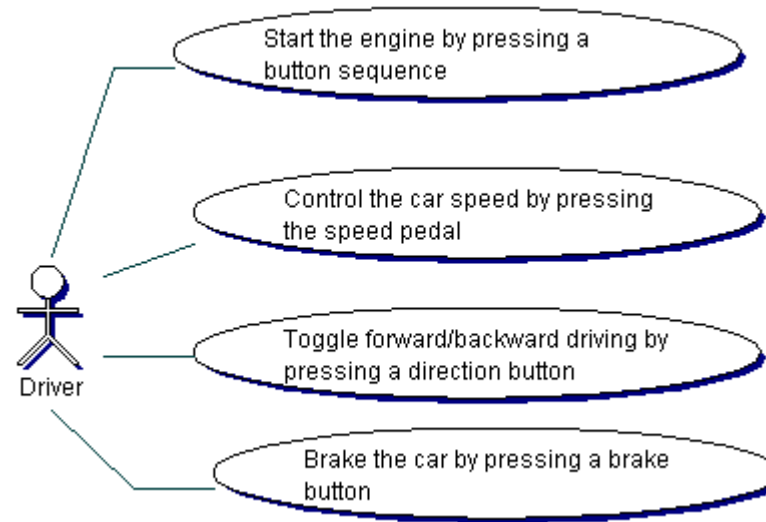
Switch to the ISO26262

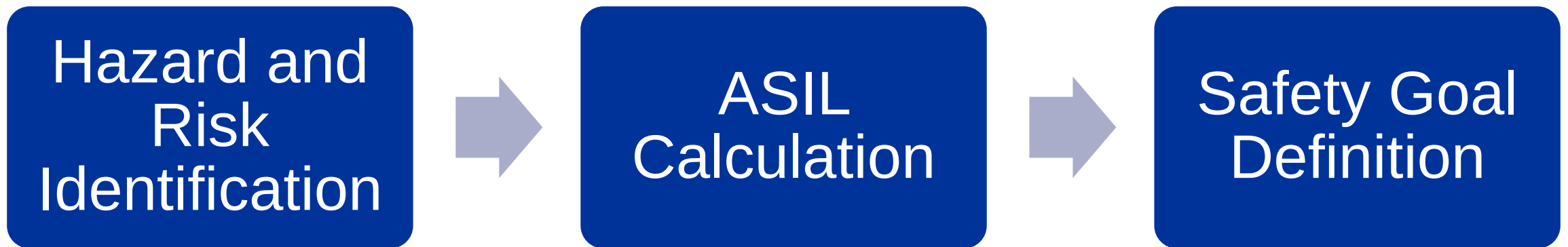
- Other than the IEC 61505, which is a type A standard, the ISO26262 is an automotive specific standard.
- Our Safety Function will be the development of an Electronic Throttle Control Function, which belongs to the automotive domain



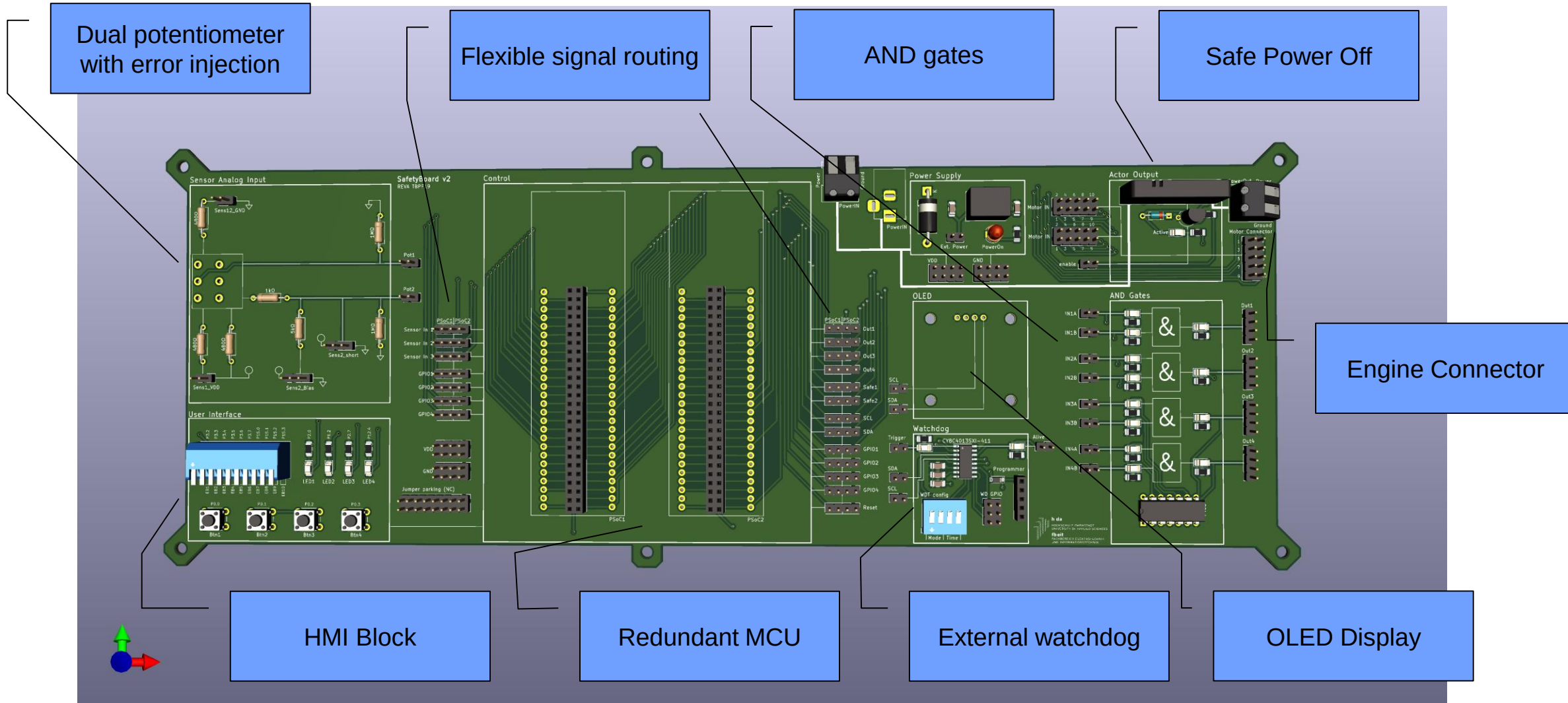
Item Description

We want to develop an electronic throttle control, providing the following features:





Our car simulation

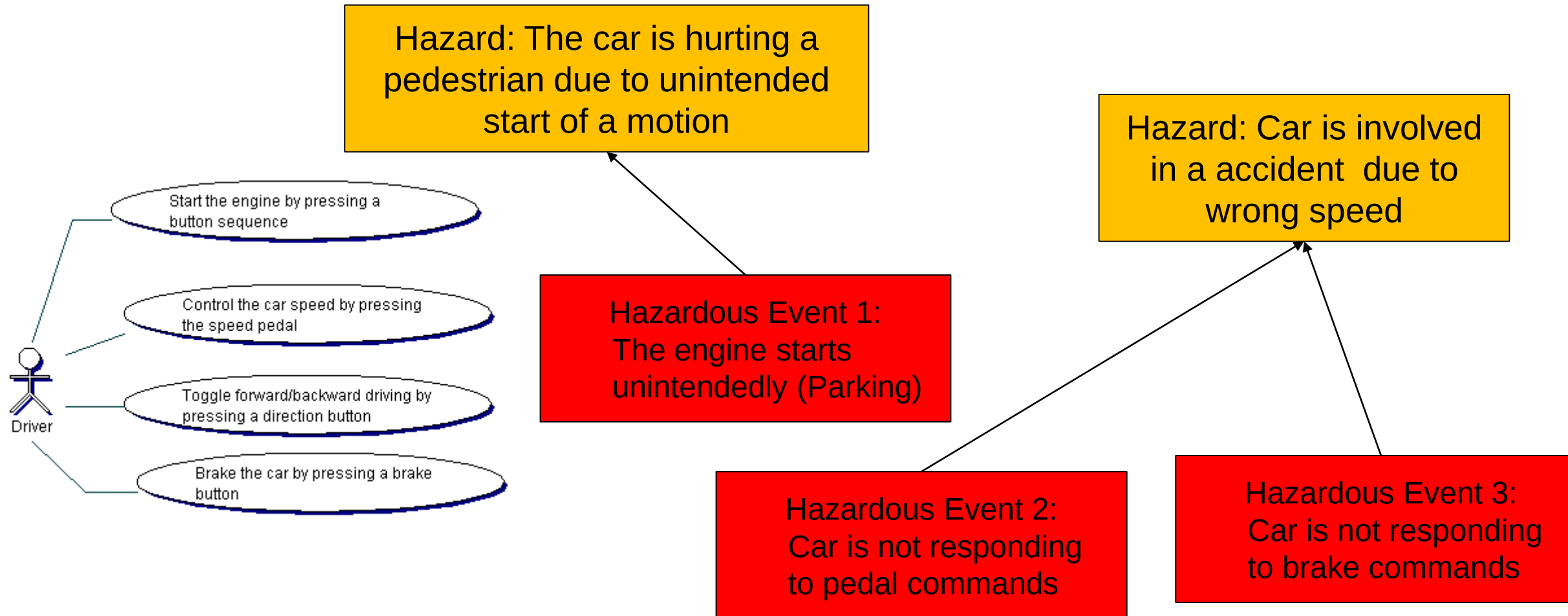


Hazard and Risk Analysis

Some definitions (based on the ISO 26262)

- Hazard - potential source of *harm* (3.74) caused by *malfunctioning behaviour* (3.88) of the *item* (3.84)
- Harm - physical injury or damage to the health of persons
- Malfunctioning behaviour - *failure* (3.50) or unintended behaviour of an *item* (3.84) with respect to its design intent
- Item - system (3.163) or combination of systems (3.163), to which ISO 26262 is applied, that implements a function or part of a function at the vehicle level
- Hazardous event - combination of a *hazard* (3.75) and an *operational situation* (3.104)
- Operational situation - scenario that can occur during a vehicle's life
- Risk - combination of the probability of occurrence of *harm* (3.74) and the *severity* (3.154) of that *harm* (3.74)

Let's try to find some hazards and hazardous events



Hazardous Event Evaluation (ISO26262 Part 3)

Table 1 — Classes of severity

	Class			
	S0	S1	S2	S3
Description	No injuries	Light and moderate injuries	Severe and life-threatening injuries (survival probable)	Life-threatening injuries (survival uncertain), fatal injuries

Driver

Table 3 — Classes of controllability

	Class			
	C0	C1	C2	C3
Description	Controllable in general	Simply controllable	Normally controllable	Difficult to control or uncontrollable

Driving Situation

Table 2 — Classes of probability of exposure regarding operational situations

	Class				
	E0	E1	E2	E3	E4
Description	Incredible	Very low probability	Low probability	Medium probability	High probability

Severity class	Exposure class	Controllability class		
		C1	C2	C3
S1	E1	QM	QM	QM
	E2	QM	QM	QM
	E3	QM	QM	A
	E4	QM	A	B
S2	E1	QM	QM	QM
	E2	QM	QM	A
	E3	QM	A	B
	E4	A	B	C
S3	E1	QM	QM	A ^a
	E2	QM	A	B
	E3	A	B	C
	E4	B	C	D

Severity

Table B.1 — Examples of severity classification

	Class of severity (see Table 1)			
	S0	S1	S2	S3
Description	No injuries	Light and moderate injuries	Severe and life-threatening injuries (survival probable)	Life-threatening injuries (survival uncertain), fatal injuries
Reference for single injuries (from AIS scale)	AIS 0 and less than 10 % probability of AIS 1-6; or damage that cannot be classified safety-related	More than 10 % probability of AIS 1-6 (and not S2 or S3)	More than 10 % probability of AIS 3-6 (and not S3)	More than 10 % probability of AIS 5-6
Examples	— Bumps with roadside infrastructure — Pushing over roadside post, fence, etc. — Light grazing damage — Damage entering/exiting parking space — Leaving the road without collision or rollover	— Side impact with a narrow stationary object, e.g. passenger car crashing into a tree (impact to passenger cell) with very low speed — Rear/front collision with another passenger car with very low speed — Front collision (e.g. rear-ending another vehicle, semi-trailer, etc.) without passenger compartment deformation	— Side impact with a narrow stationary object, e.g. passenger car crashing into a tree (impact to passenger cell) with low speed — Rear/front collision with another passenger car with low speed — Pedestrian/bicycle accident with low speed	— Side impact with a narrow stationary object, e.g. passenger car crashing into a tree (impact to passenger cell) with medium speed — Rear/front collision with another vehicle with medium speed — Front collision (e.g. rear-ending another vehicle, semi-trailer, etc.) with passenger compartment deformation
NOTE The informative examples in Table B.1 can be applied to passenger cars and T&B, but are considered on a case by case basis.				

Exposure (considering duration in operational situations)

Table B.2 — Classes of probability of exposure regarding duration in operational situations

	Class of probability of exposure in operational situations (see Table 2)			
	E1	E2	E3	E4
Description	Very low probability	Low probability	Medium probability	High probability
Duration (% of average operating time)	Not specified	<1 % of average operating time	1 % to 10 % of average operating time	>10 % of average operating time
Examples for road layout	—	— Country road intersection — Highway exit ramp	— One-way street (city street)	— Highway — Country road

Table B.2 (continued)

	Class of probability of exposure in operational situations (see Table 2)			
	E1	E2	E3	E4
Examples for road surface	—	— Snow and ice on road — Slippery leaves on road	— Wet road	—
Examples for vehicle stationary state	— Vehicle during jump start — In repair garage	— Trailer attached — Roof rack attached — Vehicle being refuelled	— Vehicle on a hill (hill hold)	—
Examples for manoeuvre	— Driving downhill with engine off (mountain pass)	— Driving in reverse — Overtaking — Parking (with trailer attached)	— Heavy traffic (stop and go)	— Accelerating — Decelerating — Stopping at traffic light (city street) — Lane change (highway)

Exposure (frequency in operational situations)

Table B.3 — Classes of probability of exposure regarding frequency in operational situations

	Class of probability of exposure in operational situations (see Table 2)			
	E1	E2	E3	E4
Description	Very low probability	Low probability	Medium probability	High probability
Frequency of situation	Occurs less often than once a year for the great majority of drivers	Occurs a few times a year for the great majority of drivers	Occurs once a month or more often for an average driver	Occurs during almost every drive on average
Examples for road layout	—	— Mountain pass with unsecured steep slope	—	—
Examples for road surface	—	— Snow and ice on road	— Wet road	—
Examples for vehicle stationary state	— Stopped, requiring engine restart (at railway crossing) — Vehicle being towed	— Roof rack attached	— Vehicle being refuelled — Vehicle on a hill (hill hold)	—
Examples for manoeuvre	—	— Evasive manoeuvre, deviating from desired path	— Overtaking	— Shifting transmission gears — Executing a turn (steering) — Using indicators — Driving in reverse

Controllability

Table B.6 — Examples of possibly controllable hazardous events by the driver or by the persons potentially at risk

Description	Class of controllability (see Table 3)			
	C0	C1	C2	C3
	Controllable in general	Simply controllable	Normally controllable	Difficult to control or uncontrollable
Driving factors and scenarios	Controllable in general	More than 99 % of the average drivers or other traffic participants are able to avoid harm	Between 90 % and 99 % of the average drivers or other traffic participants are able to avoid harm	Less than 90 % of the average drivers or other traffic participants are able to avoid harm
Example situations that are considered distracting e.g. unexpected radio volume increase or warning message - fuel low	Maintain intended driving path	—	—	—
Example for unavailability of a driver assisting system that does not affect the safe operation of the vehicle	Maintain intended driving path	—	—	—
Example for unintended closing of window while driving	—	Remove arm from window	—	—
Example for blocked steering column when accelerating from standstill	—	Brake to slow/stop vehicle	—	—
Example for failure of ABS during emergency braking	—	—	Maintain intended driving path	—
Example for propulsion failure at high lateral acceleration	—	—	Maintain intended driving path	—
Example for inadvertent opening bus door while driving with passenger standing in doorway	—	—	Passenger grabs hand rail to avoid falling out of bus	—
Example for failure of brakes	—	—	—	Steer away from objects in driving path

Example argument: Engine is activated unintendedly

Severity

- As the car starts from a standstill situation, only low speeds are likely.
- Therefore only light or moderate injuries are expected.
- ➔ S1

QM

Exposure

- According to operational statistics, starting the car is less than 1% of the driving situation
- ➔ E2

Controllability

- As we assume low speeds only, we can consider that a normal driver will be able to hit the brake
- ➔ C2

Note: The severity argument might be wrong if the car is for example located in a garage and another person is standing between car and garage wall.

Therefore it might be necessary to split the hazardous event into different situations with different severity, probability (and maybe even controllability) parameters

Hazardous Event Evaluation (sample only)

Hazardous Event	S	E	C	ASIL _{req}
The engine starts unintendedly (Parking)	S1	E2	C2	QM
Car is not responding to driver commands: Car does not speed up	S2	E4	C3	C
Car is not responding to driver commands: Car does not slow down (but we still can brake)	S2	E4	C2	B
Car is not responding to driver commands: Car is speeding up without driver interaction (we still can brake)	S2	E4	C2	B
Car is not responding to driver commands: Car is slowing down without driver interaction	S1	E4	C3	A

We might add further details to the evaluation, e.g. how drastic is the speeding up / slowing down movement, what is the current speed,....

Identification of Safety Goals

- A safety goal shall be determined for each hazardous event with an ASIL evaluated in the hazard analysis and risk assessment.
- Safety goals are top-level safety requirements for the item. They lead to the functional safety requirements needed to avoid an unreasonable risk for each hazardous event.
- Safety goals are not expressed in terms of technological solutions, but in terms of **functional objectives**.
- If similar safety goals are determined, these may be combined into one safety goal.
- The ASIL determined for the hazardous event shall be assigned to the corresponding safety goal. If similar safety goals are combined into a single one, in accordance with 6.4.4.1, the highest ASIL shall be assigned to the combined safety goal.

HAZOP method

HAZOP stands for HAZARD AND OPERABILITY and aims to identify critical operational modes using a keyword based approach.

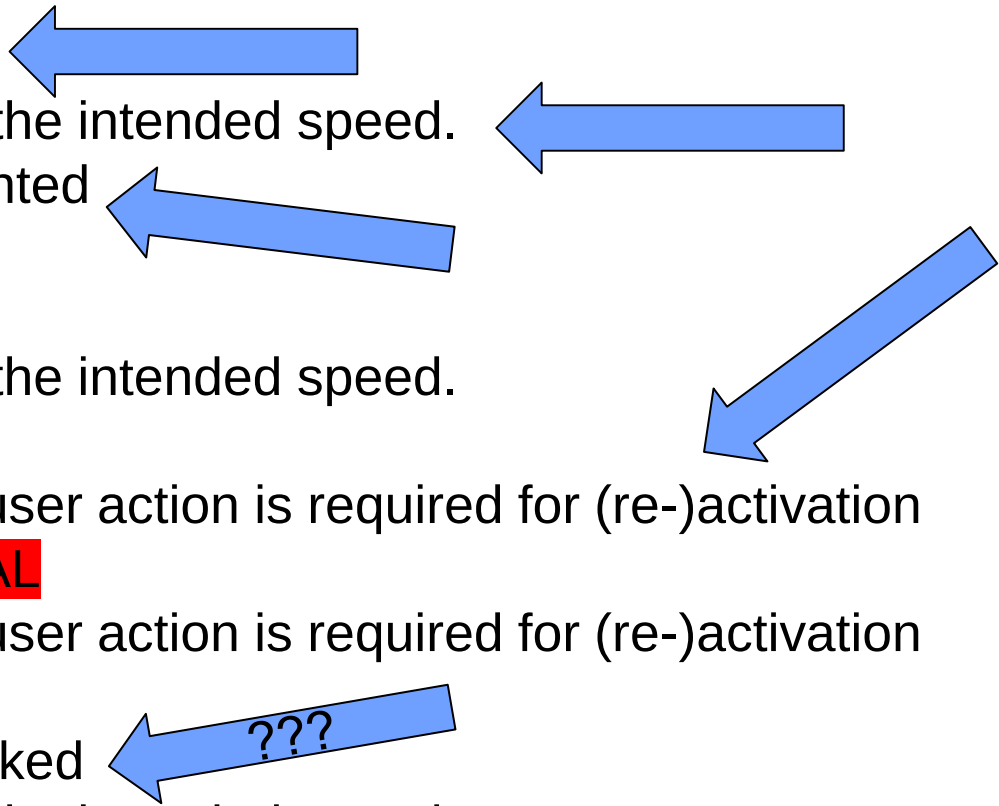
- Hazard: Unintended movement of the engine may harm humans

What means unintended?

- Too fast?
- Too slow?
- Not at all?
- Starting at the wrong moment / state?
- Instable / oscillating?

Development of Safety Goals based on HAZOP analysis

Hazard: Unintended movement of the engine may harm humans

- Too fast? **CRITICAL**
 - The targetspeed signal must be reliable
 - The engine speed must be verified against the intended speed.
 - A max speed supervision must be implemented
 - Too slow? **Depending on situation**
 - The targetspeed signal must be reliable
 - The engine speed must be verified against the intended speed.
 - Not at all? **Depending on situation**
 - If the engine has come to a halt, a specific user action is required for (re-)activation
 - Starting at the wrong moment / state? **CRITICAL**
 - If the engine has come to a halt, a specific user action is required for (re-)activation
 - Instable / oscillating?
 - The engine current speed needs to be checked
 - The engine speed must be verified against the intended speed.
- 

Safety Requirements

The functional safety requirements shall be derived from the safety goals, **considering the system architectural design**.

The functional safety requirements shall specify, if applicable, strategies for:

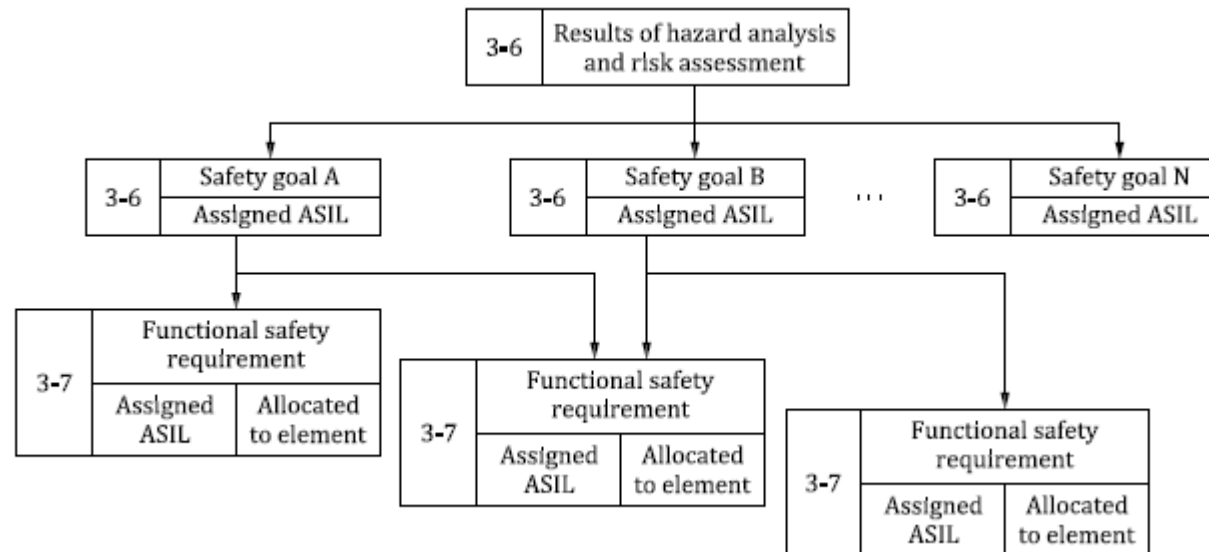
- a) fault avoidance;
- b) fault detection and control of faults or the resulting malfunctioning behaviour;
- c) transitioning to a safe state, and if applicable, from a safe state;
- d) fault tolerance;
- e) the degradation of the functionality in the presence of a fault and its interaction with f) or g);
EXAMPLE Maintaining the vehicle in a limp-home mode until the ignition has been switched from "on" to "off".
- f) driver warnings needed to reduce the risk exposure time to an acceptable duration;
- g) driver warnings needed to increase the controllability by the driver (e.g. engine malfunction indicator lamp, ABS fault warning lamp);
- h) how timing requirements at the vehicle level are met, i.e. how the fault tolerant time interval shall be met by defining a fault handling time interval;

Safety Requirements (cont.)

Each functional safety requirement shall be specified by considering the following, as applicable:

- a) operating modes;
- b) fault tolerant time interval;
- c) safe states;
- d) emergency operation time interval; and
- e) functional redundancies (e.g. fault tolerance).

Safety Goals and Safety Requirements



Fault / Error / Failure

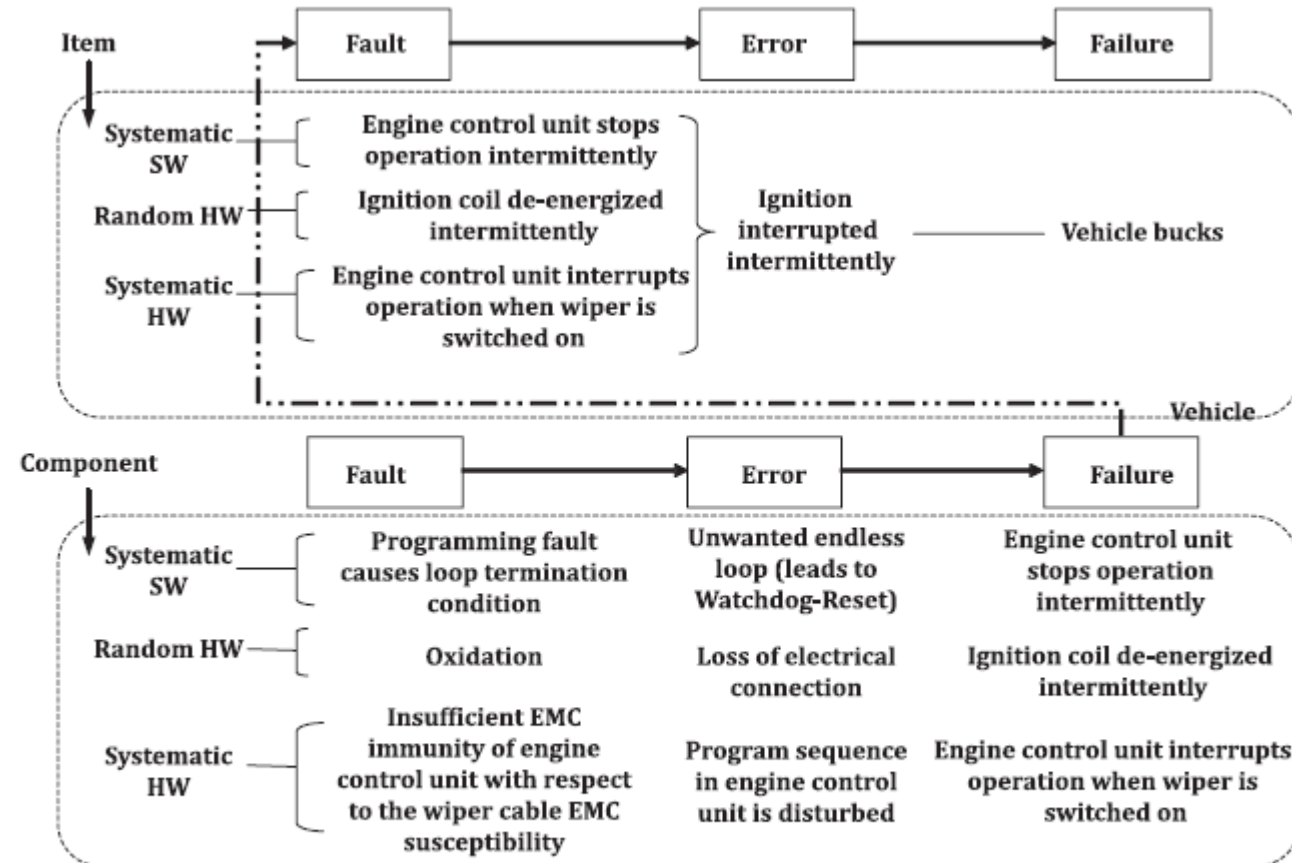


Figure 5 — Example of faults leading to failures

Fault Tolerant Time Interval

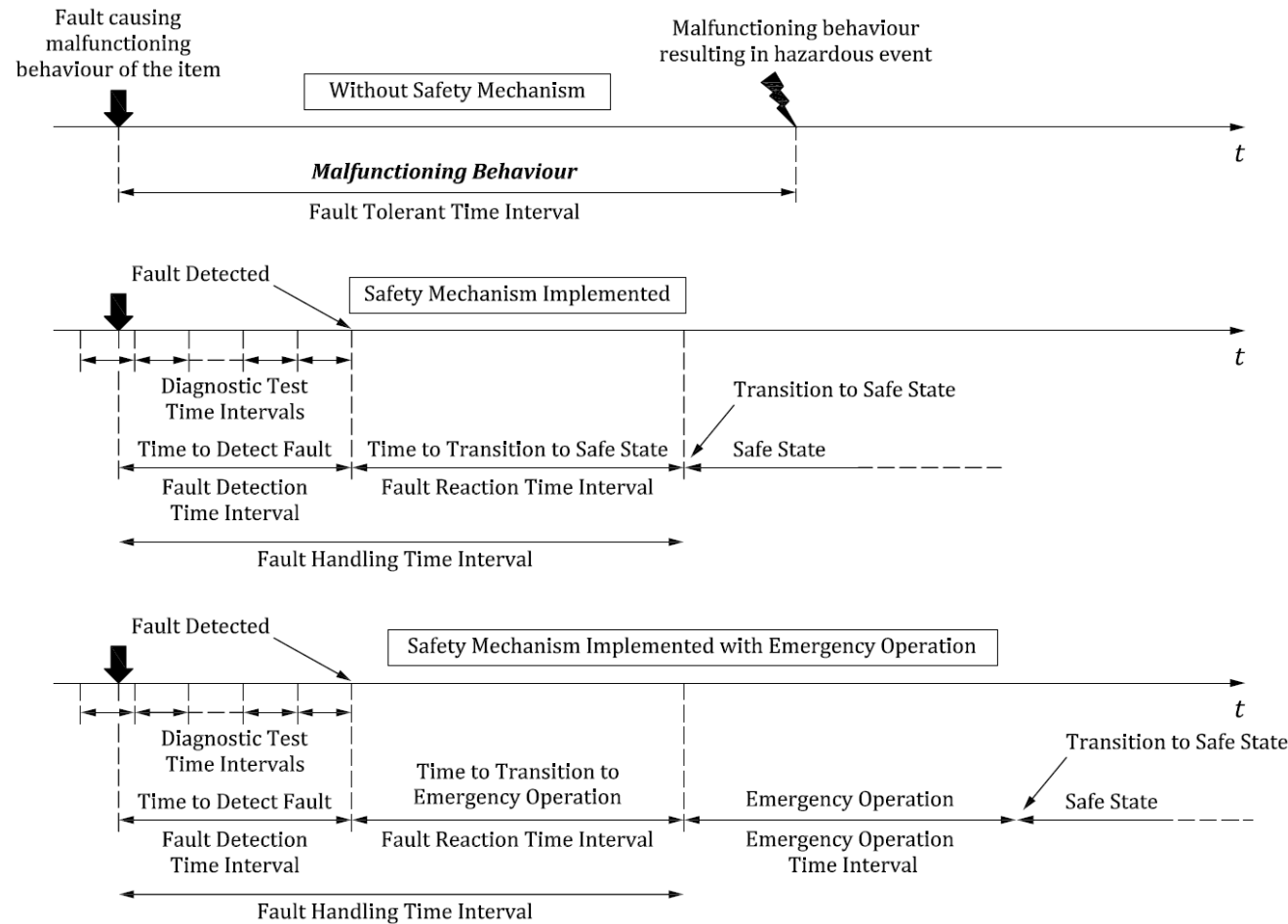


Figure 5 — Safety relevant time intervals

No safety function

Failsafe

Failoperational

Example Safety Goal and Requirements: Gas Pedal Position Reading Error

Hazardous Event

- Gas Pedal Position Reading Error

Safety Goals

- The gas pedal signal must be reliable, wrong values must be detected

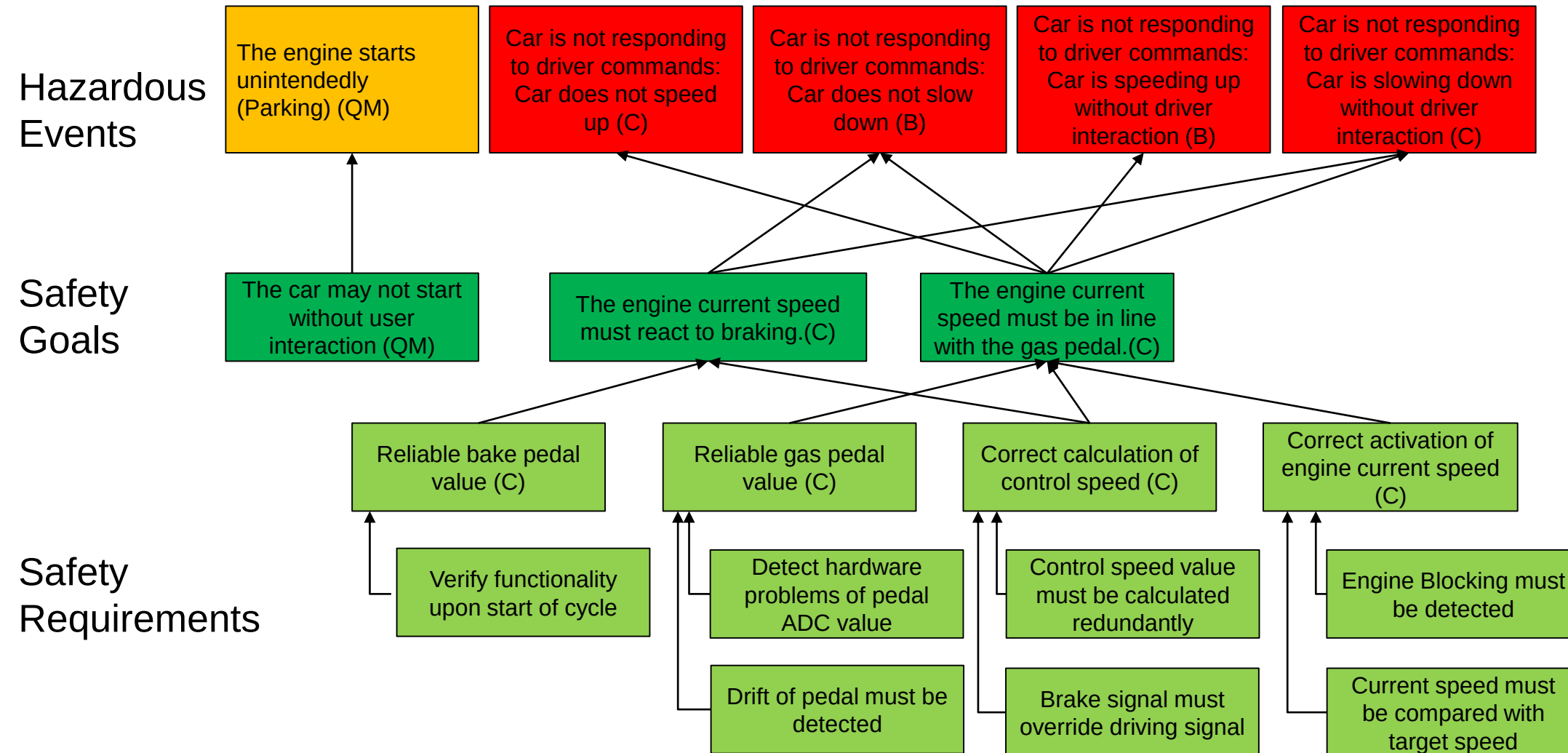
Safety Requirements

- Pedal position will be read by redundant sensors
- Detect hardware problems (broken connections, bad connections, short circuits) of pedal ADC value by adding additional resistors

Example Safety Goal and Requirements: Gas Pedal Position Reading Error

	Pedal position must be read by redundant sensors	Detect hardware problems (broken connections, bad connections, short circuits) of pedal ADC value
a) fault avoidance;	Not possible	Not possible
b) fault detection and control of faults or the resulting malfunctioning behaviour;	The sensor value will be read by 2 different controllers (inverted signals). The measured data will be compared.	Resistors will be used to detect short circuit and broken connections
c) transitioning to a safe state, and if applicable, from a safe state;	Up to 3 values may be different. Afterwards transition to slow speed and stop.	Car will go down to slow speed mode, only considering the correct sensor value. In case both sensors report error, car will stop.
d) fault tolerance;	<10% ok <25% slow speed Else stop	As given by resistors
e) the degradation of the functionality in the presence of a fault	Speed will be slowed down or stopped	Speed will be slowed down or stopped
f) driver warnings needed to reduce the risk exposure time to an acceptable duration;	Not applicable	Not applicable
g) driver warnings needed to increase the controllability by the driver (e.g. engine malfunction indicator lamp, ABS fault warning lamp);	Warning will be shown in display in case of slow down mode	Warning will be shown in display
h) how timing requirements at the vehicle level are met, i.e. how the fault tolerant time interval shall be met by defining a fault handling time interval;	ADC values will be measured every 10ms. Engine speed will be updated on validated data every 50ms.	ADC values will be measured every 10ms. Engine speed will be updated on validated data every 50ms.

Overview of a subset of Safety Goals and Safety Requirements (first draft)



Last Lecture: Design and Implementation

- The hazard and risk analysis tells us which SIL level we require
- Now we only have to figure out which SIL our system actually has – and improve it when required

Hardware

- Reliability of components
- Hardware design
 - Redundancy
 - Common cause failure analysis
 - Freedom from interference
- Verification and validation / FMEDA / FMA
- ...

Part 2

Software

- Reliability of development process
- Software design
 - Diagnostic coverage
 - Freedom from interference
- Validation and Verification
- ...

Part 3

What do we have?

- Potential hazards and risks (hazardous events) leading to possible harms have been identified.
- High level safety goals and safety requirements have been specified.
- The required ASIL has been calculated based on the identified hazardous events and has been mapped to the Safety Goals.

During this process, we have taken decisions based on experience data and assumptions of use of the item. Make sure to document and review them!

In the next step, we have to develop a safety architecture and check, if this architecture fulfills the specified requirements and ASIL levels.